

Open Switch Fault-Tolerant VOC-PI Controller based Vienna Rectifier for EV Charging Stations

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Abstract—Power electronic converters are one of the key facets of modern electric vehicle (EV) charging applications. In the last few years, many EV charging manufacturers have been convinced that robust power converters and fault tolerance are essential for successful EV adoption, and cost-effective and simple fault tolerance techniques are necessary to achieve widespread EV use. The rise in the input current harmonics leads to a poorer power factor on the grid side during an open switch fault condition. The proposed controller provides better performance despite the open-switch fault occurring in any phase. The proposed voltage-oriented controller with Vienna rectifier helps to reduce the input current THD to less than 5% and unity power factor at AC mains, which meets the IEEE-519 standards.

Keywords— Vienna rectifier, voltage oriented controller, fault detection, electric vehicles, charging stations

I. INTRODUCTION

Vienna rectifier has been widely used in various applications such as EV charging stations, welding power sources, and aircraft applications [1-5]. The high performance, low losses, and simplification control strategy of Vienna rectifier topology have attracted major attention from industry and academia [1]. The Vienna rectifier serves the key features of mitigating the input current THD and improves the power factor at AC mains for the medium and high power applications. However, the reliability of power converters with the increase in switching operations with high frequency and power density as malfunctions are detrimental to the system operation [6]. During the charging of electric vehicles, the three-phase Vienna rectifier suffers a high-frequency voltage shock owing to temperature fluctuations, overload, and inaccurate gate driving signals [7]. A study shows that between 2000 – 2004, Sweden's wind farms failed, out of 17.5% failure, 14.3% of power converter inaction times [8]. The power semiconductor switches frequently malfunctions compared to other components in the system contributing to 31% from system malfunctions

compared to the other sensitive components such as capacitors, resistors, gate drivers, and inductors [8]. The most common sources of power conversion failures in industrial applications such as wind energy systems, aircraft applications, telecommunication power supplies, and EV charging stations are 21% failure from power semiconductor switches and 13% failure from solders. In comparison, 26% of PCB (Printed Circuit Board) and 30% of capacitors are the main concerns in a recent study of more than 200 products from 80 companies [8]. Power switch malfunction is classified into open-switch fault and short-switch fault. Current flowing in the system due to the short-switch faults result in system shutdown. This short-switch fault can be avoided using external hardware approaches such as fuses and circuit breakers [9]. In Vienna rectifier, it is very difficult to control the power flow in the system when a short circuit occurs in one switch within the same phase. When a short circuit occurs in one switch with a positive current, the gate signal controls the other switch in the same phase. However, it is challenging to control the operation of the system when the current flow is negative. The system will nonetheless not be shut down by the open switch fault, but rather it will decrease the system performance and create current distortion in the system. In [10], using two tolerance control technique to identify and diagnose the system fault makes the control system more difficult. In [11], a new controller has been developed to analyze and diagnose the fault in the system. In [12], the three-phase fault diagnosis method diagnoses the fault in the wind energy-based converter system. Various fault diagnosis methods with the use of additional devices [9, 13-15]. The PFC controller with Vienna rectifier for the EV charging station is proposed in [16]. Harmonics in the PFC controller-based Vienna rectifier is more than 5% during fault. In [7], the Vienna rectifier with fault-tolerant control has been proposed. In the existing system, the input current THD is more than 5% which could not be able to meet the IEEE-519 standards. In this study, the proposed system is simulated and compared with existing system. The prototype

of Vienna rectifier with PWM rectifier available in the market is shown in Fig. 1.

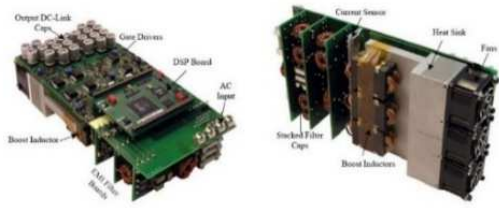
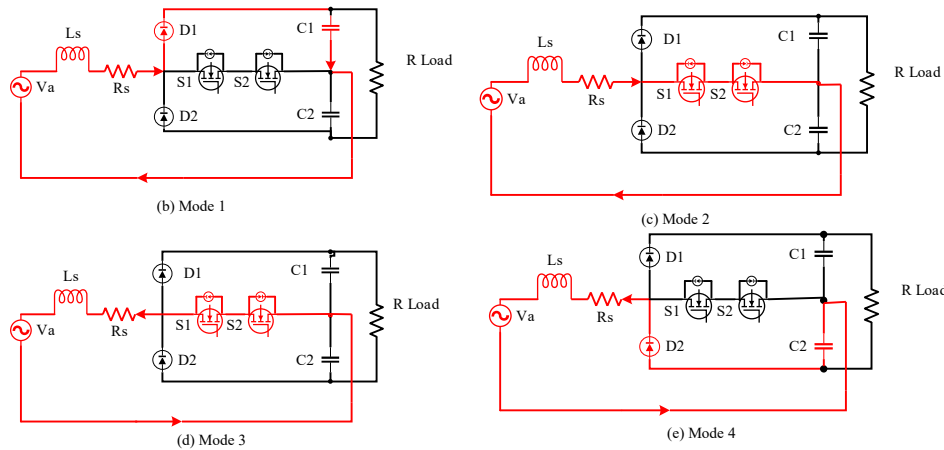
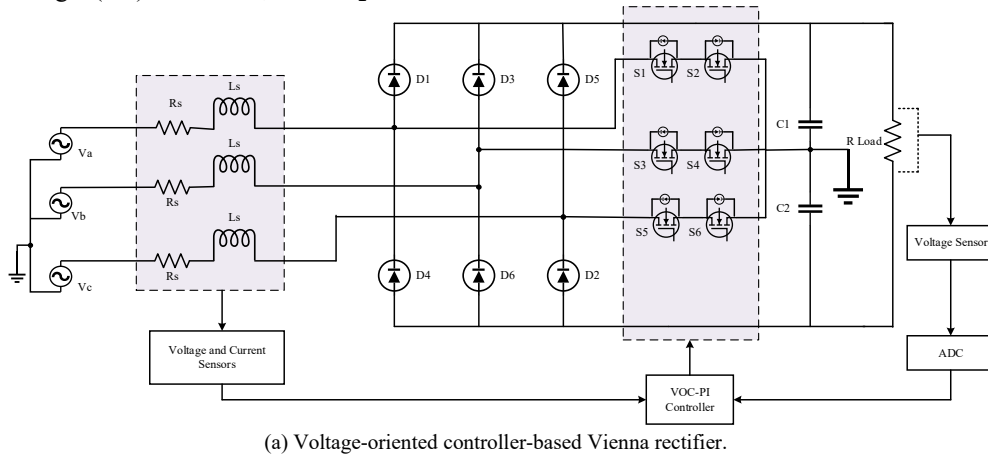


Fig. 1. Commercially available air-cooled Vienna Rectifier with PWM switching

II. VIENNA RECTIFIER

The Vienna rectifier with voltage oriented controller is shown in Fig. 2. On every diode and semiconductor switch, the voltage stress is $V_{dc}/2$. The current flow in one leg of Vienna rectifier is shown in Fig. 2(b-e). In mode 1, diode D_1 conducts

for the positive (half cycle) reference voltage while the power semiconductor switches are turned off. The current flow during this period is $V_a L_s R_s D_1 C_1$ as shown in Fig. 2(b). The current flows through $V_a L_s R_s S_1 S_2$ for the positive reference voltage with semiconductor switches in ON condition, as shown in Fig. 2(c). Meanwhile, for the reference voltage with semiconductor switches in ON and OFF conditions, the current flows through $S_1 S_2 R_s L_s V_a$ and $C_2 D_2 R_s L_s V_a$, respectively. The current flow during the negative reference voltage with semiconductor switches in ON and OFF conditions are shown in Fig. 2(d) and Fig. 2(e), respectively. In addition, a conventional controller with Vienna rectifier is demonstrated in Fig. 3. The output voltage of less than 200V DC for the three-phase input supply is one of the major drawbacks of the conventional controller, applicable only for slow charging stations (level 1) for EVs [16].



(b-e) Single Phase operations of the controller

Fig. 2. VOC-PI Vienna Controller

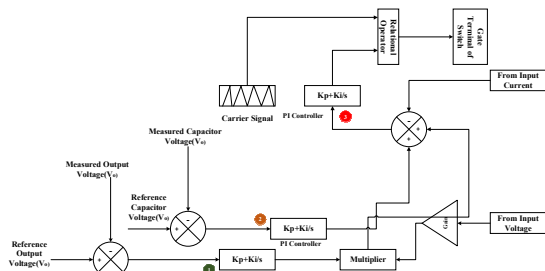


Fig. 3. Conventional PFC controller

III. VIENNA RECTIFIER WITH OPEN-SWITCH FAULT

When the open-switch fault occurs in the system, the voltage level and current levels will be reduced. As a result, harmonics will be generated in the input current during the open-switch fault. This will lead to a poor power factor on the grid side. In this paper, the VOC-PI Controller has been proposed to mitigate the grid voltage and current imbalance and mitigates the input current THD, which improves the system performance even during the open-switch fault condition. The power electronic converter's operation is

highly reliant on the control structure that has been established. The voltage-oriented controller helps to mitigate many issues in the power electronic converters. The voltage controller and current controller present in the VOC-PI controller helps to mitigate the voltage ripples at DC output voltage and input current harmonics at AC mains, respectively [17]. In the theoretical platform, most commonly used power controller is a voltage-oriented controller, which provides a high input power factor, less input current THD, and ripple-free DC output voltage to control the charging mechanism for EVs, as shown in Fig. 2. [18]. The Equation (1) and (2) demonstrates the Clark and Park transformation matrices for VOC technique. The control technique of the voltage-oriented controller with the PWM technique is shown in Fig. 4.

$$\begin{bmatrix} v_{sa} \\ v_{sb} \\ v_0 \end{bmatrix} = \sqrt{\frac{2}{3}} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} v_{sa} \\ v_{sb} \\ v_{sc} \end{bmatrix} \quad (1)$$

$$\begin{bmatrix} v_d \\ v_q \end{bmatrix} = \begin{bmatrix} \sin\theta & \cos\theta \\ -\cos\theta & \sin\theta \end{bmatrix} \begin{bmatrix} v_{sa} \\ v_{sb} \end{bmatrix} \quad (2)$$

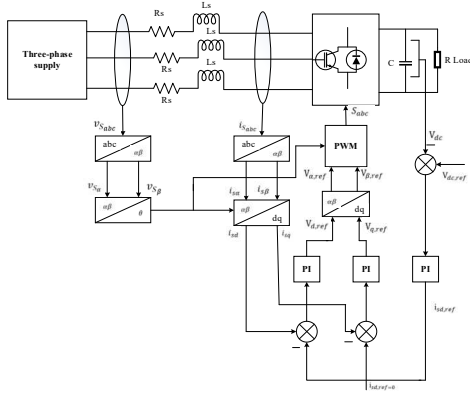


Fig. 4. Control Technique for Vienna rectifier

The following approach helps to mitigate the steady-state errors using PI controller [19]:

$$v_{d,ref} = K_p(i_{sd,ref} - i_{sd}) + K_i \int (i_{sd,ref} - i_{sd}) dt \quad (3)$$

$$v_{q,ref} = K_p(i_{sq,ref} - i_{sq}) + K_i \int (i_{sq,ref} - i_{sq}) dt \quad (4)$$

where, K_p and K_i is the gain of the proportional-integral controller, input current in dq0 domain are i_{sd} and i_{sq} , and $i_{sd,ref}$ and $i_{sq,ref}$ are the reference signals for i_{sd} and i_{sq} .

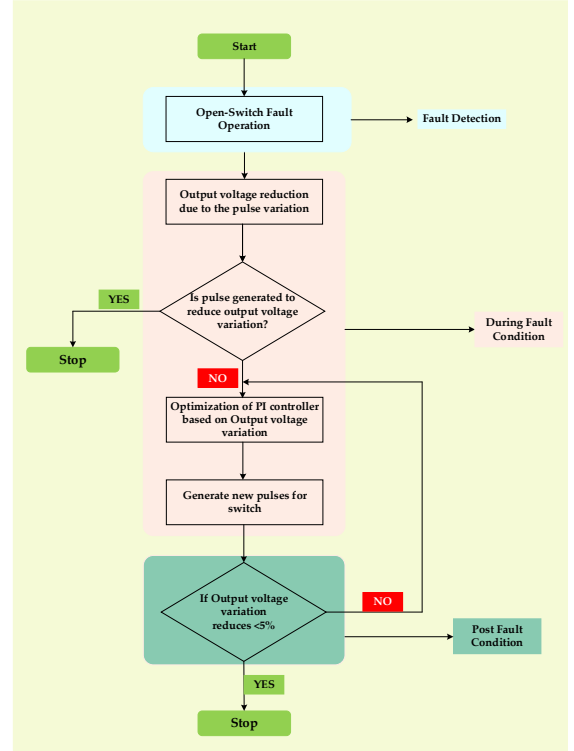


Fig. 5. Block diagram of fault diagnoses of Vienna rectifier

IV. SIMULATION RESULTS

The overall methodology of fault detection technique is illustrated in Fig. 5. The voltage-oriented controller with Vienna rectifier is modeled using MATLAB/Simulink. The total number of controller switches in the Vienna rectifier is six (two in each phase with a series connection). In addition, one phase (Phase A) is disconnected from the circuit to analyze the fault tolerance in open switch conditions.

TABLE I. SPECIFICATION OF VOLTAGE-ORIENTED CONTROLLER WITH VIENNA RECTIFIER

Parameter	Symbol	Unit	Value
Resistive Load	R_{load}	Ω	15
Input inductance	L_f	mH	5
Grid frequency	f	Hz	50
Switching frequency	f_s	kHz	12
Input resistance	R_s	Ω	5
Output Power	P_{out}	kW	21.66
Input voltage	V_{in}	V	400
Output voltage	V_{dc}	V	570
Current controller	K_{p1} and K_{p2}	-	67
Current controller	K_{i1} and K_{i2}	-	1.7×10^4
Voltage controller	K_{p3}	-	0.00027
Voltage controller	K_{i3}	-	0.017

The three-phase current waveforms during the open switch fault condition for phase A are shown in Fig. 6. It shows that when the open switch fault occurs in one phase leads to voltage/current distortion in the input source side. Fig. 7 shows the post-open switch fault of input current for the transient condition. It demonstrates that the magnitude of the open-switch fault phase is reduced to 22.2% compared to the other two phases. The input current harmonics phase A, phase B, and phase C post-open-switch fault in Phase A are shown in Fig. 8, Fig. 9, and Fig. 10, respectively. The input current THD of three-phase after fault condition is 4.19%, 3.59%, and 3.92%, which is less than 5% to meet the IEEE-519

standards. The output voltage is shown in Fig. 11. it can be seen from Fig.11 the ripples in the output voltage are small and negligible. Therefore, the output voltage from Vienna rectifier is nearly 600 V DC which is most suitable for EV charging stations.

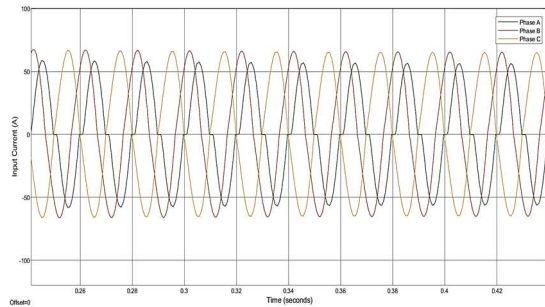


Fig. 6. Three-phase input current during open-switch occurred in Phase A

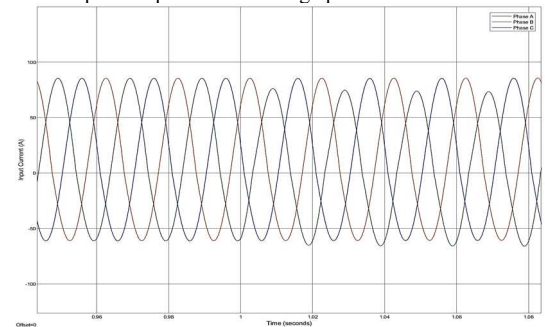


Fig. 7. Post open-switch fault of input current for the transient condition.

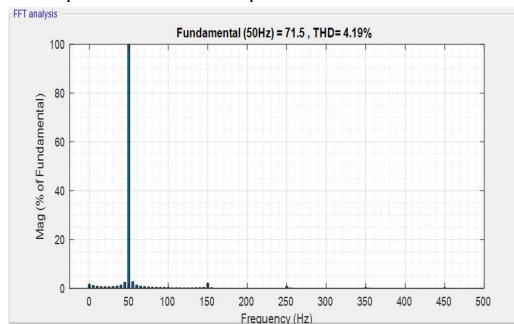


Fig. 8. The input current THD of Phase A post-open-switch fault occurred in Phase A.

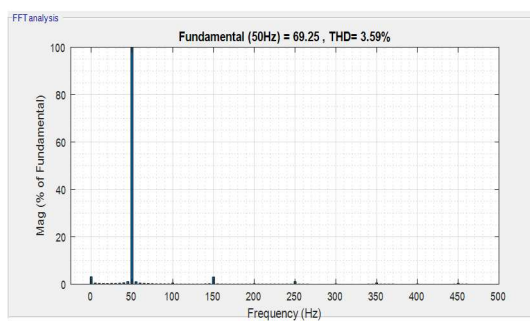


Fig. 9. The input current THD of Phase B post-open-switch fault occurred in Phase A.

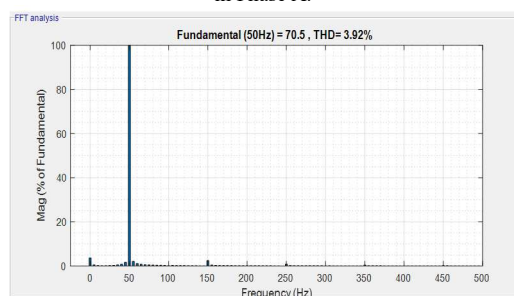


Fig. 10. The input current THD of Phase C post-open-switch fault occurred in Phase A.

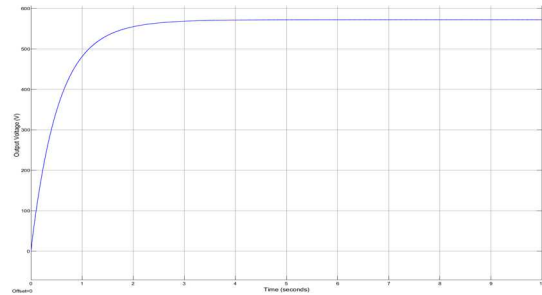


Fig. 11. DC Output Voltage

V. DISCUSSIONS

The simulation results demonstrate that the magnitude of the faulted phase acts as a conventional diode bridge rectifier system with a 22.2% reduction in the phase current compared to the other two phases under normal conditions. The harmonics in the input current of Phase A, Phase B, and Phase C is 4.19%, 3.59%, and 3.92%, respectively. This demonstrates that the harmonics at each phase are less than 5% during the open-switch fault, which meets the IEEE-519 standards. As a result, the PI controller is optimized utilizing the VOC controller outperforms the conventional PFC controllers for EV charging stations. The comparison of input current THD for PFC controller and VOC-PI controller has been tabulated in Table II. Results show that the input current harmonics during the tolerant control condition for the conventional PFC controller in each phase is 25.2%, 16.4%, and 12.5% [7], whereas the harmonics in the VOC-PI controller with Vienna rectifier is an average of 3.8% which can satisfy the IEEE-519 standards. Based on Table II, it can be seen, there is an improvement in input current harmonics of an average of 18% compared to the PFC controller.

VI. CONCLUSION

In this paper, the fault-tolerant control for the Vienna rectifier is proposed for EV charging stations, and the Vienna rectifier with VOC controller is modeled using MATLAB/Simulink software. Furthermore, the open-switch fault for Phase A has been conducted, and input and output parameters such as input current THD and output voltage have been analyzed, and the results are compared with the existing conventional PFC controller. The findings indicate that the Vienna rectifier VOC-PI controller for EV charging stations alleviates system failures, reduces the input current THD to less than 5% in each phase, and meets the IEEE-519 standards. Hence, when the open-switch fault occurs in any one switch in each phase. Thus, it can be seen, the VOC-PI controller with Vienna rectifier for an electric vehicle charging station improves the performance 17.7% better compared to a conventional PFC controller.

TABLE II. COMPARISON OF INPUT CURRENT THD DURING OPEN-SWITCH FAULT IN EACH PHASE FOR PFC CONTROLLER AND VOC-PI CONTROLLER.

Parameter	Phase	Normal Condition		Fault Condition		Tolerant Control Condition	
		PFC	VOC-PI	PFC	VOC-PI	PFC	VOC-PI
Current THD	A	7.4%	2.32%	96.4%	95.3%	25.2%	4.19%
	B	7.4%	3.59%	95.8%	3.59%	16.4%	3.59%
	C	7.4%	3.92%	97.3%	3.92%	12.5%	3.92%

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